# Petrophysics Lecture 6: Density & Neutron Porosity Logs

Gas Effect & Lithology Crossplots

**Department / Year: Petroleum Geosciences Department - 2nd Year**

## Lecture Key Outlines & Core Competencies:

- 1. Bulk Density Measurement and Matrix Density Selection
- 2. Thermal Neutron Log and Hydrogen Index
- 3. Density-Neutron Crossover for Gas Zone Detection

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## Section 2: Practical Calculations, Laboratory Equations & Exercises

### Core Equations & Petrophysical / Geological Relations:

Equation 1: Phi_total = (V_bulk - V_grain) / V_bulk * 100%

Equation 2: Q = (k * A * delta_P) / (mu * L)  [Darcy Law 1D]

Equation 3: Sw = ((a / Phi^m) * (Rw / Rt))^(1 / n)  [Archie Saturation]

Equation 4: V_shale = (GR_log - GR_clean) / (GR_shale - GR_clean)  [Linear Index]

## Laboratory Work Checklist & Submission:

[OK] Step 1: Calibrate electronic calipers and weigh dry clean core plugs.

[OK] Step 2: Record helium injection pressure and stabilization chamber readings.

[OK] Step 3: Plot data on semi-log scale graph and determine trend line slopes.

[OK] Step 4: Upload your lab report directly via PETROGEO Admin & Student portal.